

MECHANICAL PROPERTIES OF 2D C/SiC COMPOSITES WITH TWO DIFFERENT CARBON TOWS

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ABSTRACT

The mechanical properties of woven cloth with two sizes of carbon tows reinforced silicon carbide composites were investigated by tension, shear and bending tests. The experimental results show that the composite fabricated with small tows possesses greater stiffness and strength than the composite with large tows. Unequal tow size caused differences of the warp waviness and the void volume fraction in the composites, which led to differences in mechanical properties of composites.

Keywords: Ceramic matrix composites; Carbon fibre; Mechanical properties; Tow sizes

1. INTRODUCTION

Carbon-fibre-reinforced SiC-matrix composites (C/SiC) have been widely investigated as high-temperature structural materials because of their low density, non-linear mechanical behaviour, enhanced fracture toughness and impact resistance [1, 2]. Especially, the mechanical properties of C/SiC composites can be retained at high temperatures and under severe service environments. Up to now, Chemical Vapour Infiltration (CVI) process, especially isothermal CVI (I-CVI) process, has been proved to be a most promising process for fabricating the composites. The primary advantages of this process are the abilities to manipulate and modify the microstructure of the matrix, to tailor the fibre/matrix interface, and to fabricate complex net or near-net shaped components at relatively low temperatures [3, 4].

In the fabrication of ceramic matrix composites, the preform is of great importance, which determines the composite microstructure and micro-stress distribution, and consequently determines the composite properties and its applications. So most previous efforts have been devoted to investigation on mechanical properties of the composites with different preforms, such as 2D, 2.5D, or 3D preforms [5-7]. However, there are few studies dealing with changes of tow itself (for example, different fibre types and

tow sizes). Moreover, most studies were mainly focused on polymer matrix composites, and no description of mechanical response of the C/SiC composites with different tow sizes can be found in the recent literature. It is well known that fine tows are relatively expensive and this is particularly true for the carbon tows currently available on the market. Therefore, the properties of composites fabricated with different sizes of tows must be characterized and a considerate trade-off must be achieved between the properties and the manufacturing cost in the commercialization of these composites.

The paper here provides experimental results on the mechanical response of 2D C/SiC composites fabricated by chemical vapour infiltration with different fibre tow size. The mechanical properties and fracture behaviour of the two composites were evaluated and compared, and efforts were made to correlate the fibre tow size to the mechanical response of the composites.

2. EXPERIMENTAL

2.1 Preparation of C/SiC composite

Two different sizes of T-300 carbon fibre tows from Toray (Japan) were employed, and each tow contained 1000 filaments (denoted as 1k tow) or 3000 filaments (denoted as 3k tow). The fibre preform was

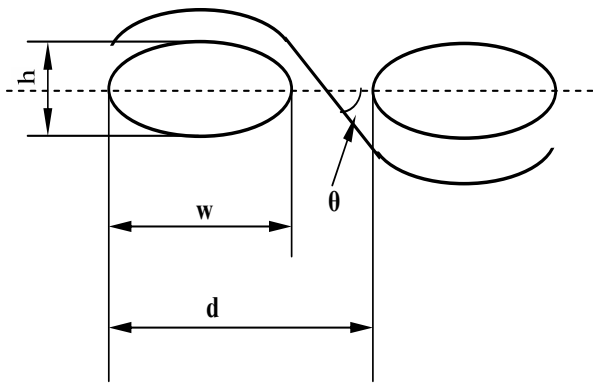


Fig.1: Schematic of cross section of plain woven cloth, θ —Undulation angle, d —Tow distance, w —Tow width, h —Tow height

prepared using a plain woven carbon-cloth, which was supplied by the Nanjing Institute of Glass Fibre, People's Republic of China. Low pressure I-CVI was used to deposit a pyrolytic carbon layer and the silicon carbide matrix. The final 2D composite fabricated with 1k carbon tows are referred to as 1k C/SiC composite, while composite with 3k carbon tows are referred to as 3k C/SiC composite. The major geometrical and other pertinent parameters of composites are presented and listed in Fig.1 and Table 1, respectively.

2.2 Measurements and observations

The tensile, three-point bending, in-plane shear tests were carried out on a universal testing machine (Instron 1196, Instron Ltd., High Wycombe, England) at room temperature. The cross-section area of gauge section of the tensile specimen for both the composites is about $3 \times 3 \text{ mm}^2$. The tensile strains were recorded with an extensometer with a gauge length of 25 mm at a crosshead speed of 0.05 mm/min. The shear deformation was measured by strain gages at a crosshead speed of 0.1 mm/min. The size of three-point bending specimens was $4 \text{ mm} \times 3 \text{ mm} \times 40 \text{ mm}$, which were tested at a crosshead speed of 0.5 mm/min with a span of 30 mm. For the respective mechanical test mentioned above, five specimens were measured for each composite, i.e., 1k C/SiC composite and 3k C/SiC composite. The fracture surfaces were observed with scanning electron microscope (SEM, HITACHI S-2700).

3. Results and discussion

3.1 Mechanical properties of composites

The static mechanical properties of the 2D C/SiC composites with 1k and 3k carbon fibre tows are listed in Table 2. Compared with the composite with 3k fibre tows, the composite with 1k fibre tows always has a higher strength in tensile, flexural, and

Table 1: Parameters of two kinds of plain woven C/SiC composites.

Parameters	1 k C/SiC	3 k C/SiC
Fiber tow count	1000 filaments per tow	3000 filaments per tow
Density/($\text{g} \cdot \text{cm}^{-3}$)	2.18(0.02)	2.07(0.03)
Fiber volume fraction/ %	40(2)	38(3)
Void volume fraction/ %	13(2)	17(2)
Undulation angle θ/ (°)	10(2)	18(4)
Tow height h/mm	0.14(0.03)	0.21(0.04)
Tow width w/mm	0.64(0.22)	1.23(0.31)
Tow distance d/mm	0.93(0.25)	1.48(0.42)

Table 2: Mechanical properties of plain woven C/SiC composites.

Properties	1 k C/SiC	3 k C/SiC
Tensile strength /MPa	312(23)	198(20)
Tensile modulus/GPa	111.3(12)	70.1(15)
Work of fracture /kJm⁻²	30.1(1.5)	13.1(2.1)
Bending strength /MPa	497(50)	441(60)
Shear strength /MPa	163(18)	127(24)

shear tests under the same testing conditions. Moreover, the tensile modulus of 1k C/SiC composite is 58% higher than that of 3k C/SiC composite, and the work of fracture (obtained from the areas under the tensile curve) of 1k C/SiC composite is more than two times that of 3k C/SiC composite. All the results show a great advantage for application of small size of tows in CMCs without any conflict in comprehensive mechanical properties.

3.2 Effects of fibre tow size on mechanical response of composites

3.2.1 Tensile properties

The tensile stress-strain curves of the two composite materials were measured up to rupture on the Instron tester at room temperature. Typical tensile stress-strain curves of the 1k and 3k C/SiC composites are presented in Fig.2. It is apparent that both the

composites behave as a typical damageable material, exhibiting an extensive non-linear stress-strain domain up to rupture because of the presence of the processing-induced microcracks and of the damage accumulation nature of materials during testing.

The curves can be generally divided into two regions: a very small initial linear region followed by a large nonlinear region. The first domain is characterized by a nearly constant elastic modulus and corresponds to partial re-opening of matrix thermal microcracks and associated partial debonding up to an applied stress of about 25 MPa; The second domain involves multiplication of the matrix microcracking and related fibre/matrix or bundle/matrix, or inter-bundle debonding, characterized by a noticeable decrease in the elastic modulus. It seems that both the composites did not show the crack saturation which is characterized by a quasi-linear region with a slight recovery of elastic modulus [7].

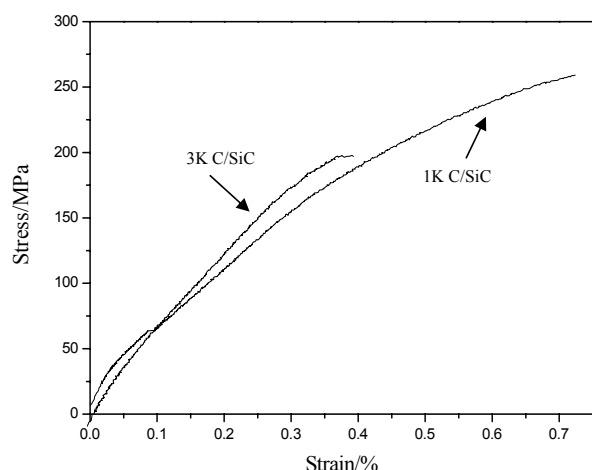


Fig.2: Typical tensile stress-strain curves of plain woven C/SiC composites.

Both the tensile curves of the composites are found to be similar. This phenomenon implies that both the composites have the same energy-dissipating mechanisms such as matrix microcracking, fibre/matrix debonding and fibre pull-out. For instance, both of the linear deformation of the 2D C/SiC composites is clearly limited up to about 25 MPa (referred to as “proportional limit” or first-matrix cracking stress σ_{mc}). However, it is interesting to notice that the composites with different tow size have different elastic modulus and strength. The 1k C/SiC composite ruptures at an average stress of 312 MPa for a cor-

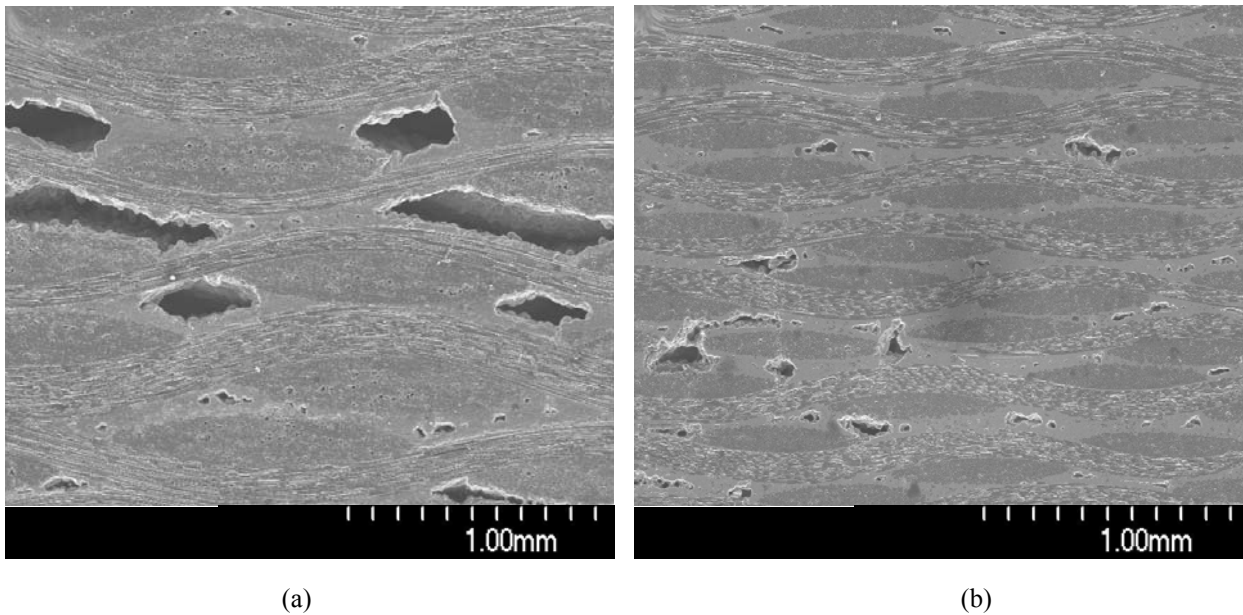


Fig.3: SEM images show the larger pores in 3k C/SiC composite than in 1k C/SiC composite.
(a) 3k C/SiC composite (b) 1k C/SiC composite

responding average strain of 0.7%, while 3k C/SiC composite at 198 MPa for 0.4%. Furthermore, the average Young's modulus obtained by the linear fitting of the initial stress-strain curves from 0 to 25 MPa is 111.3 GPa for the 1k C/SiC composite, which is about 1.5 times higher than that of 3k C/SiC composite (70.1 GPa).

Because of the larger cross-section area of 3k fibre tows, as seen from Fig.3, the 3k C/SiC composite has fewer numbers of tows than 1k C/SiC composite due to the same cross-section area. The fewer numbers of tows would have a negative effect on measured tensile strength because there is probably some slight deviation from the desired orientation or they are not centred in the tensile specimen. Moreover, larger pores are prone to be formed between fibre bundles in preform with 3k tows than in preform with 1k tows. It is difficult to fill these large pores completely in the following infiltration. In addition, the waviness issue also plays a role in poorer tensile properties since the larger tow has to undulate at a greater angle from the loading direction which would induce greater off-axis stresses on the fibres which also diminishes strength, as well as much degradation of stiffness of composites in load direction [8]. The diameter of 3k fibre tow is much larger than that of 1k fibre tow, so the waviness of tow in

3k C/SiC composite is larger than that of 1k C/SiC composite (e.g., 18° vs. 10° of undulation angle, as shown in Table 1), which possesses a non-negligible contribution to the lower elastic modulus of 3k C/SiC composite.

Both the composites show non-brittle fracture characteristics, which indicates that similar fracture mechanisms are involved in both the composites, such as matrix cracking, interfacial debonding/sliding, fibre pull-out, delamination, etc., which are described in detail elsewhere [9-12]. The lower failure strain of 3k C/SiC composite implies the limited interfacial sliding or fibre pull-out compared to 1k C/SiC composite. This is confirmed by detailed microstructure observation (Fig.4), which revealed that the matrix infiltrated in fibre tows is denser in 1k C/SiC composite, resulting in obvious fibre filaments pull-out, but for 3k C/SiC composite, the matrix in fibre tows is sparse resulting in large fibre bundles or clusters pull-out. Furthermore, the energy absorbed by clusters pull-out is much less than that by fibre filaments pull-out because of the less interfacial friction dissipation and shorter deflected crack propagation path produced by clusters pull-out. As expected, the 3k C/SiC composite manifests a lower work of fracture, with cracks rather through interlaminar paths than across the fibres.

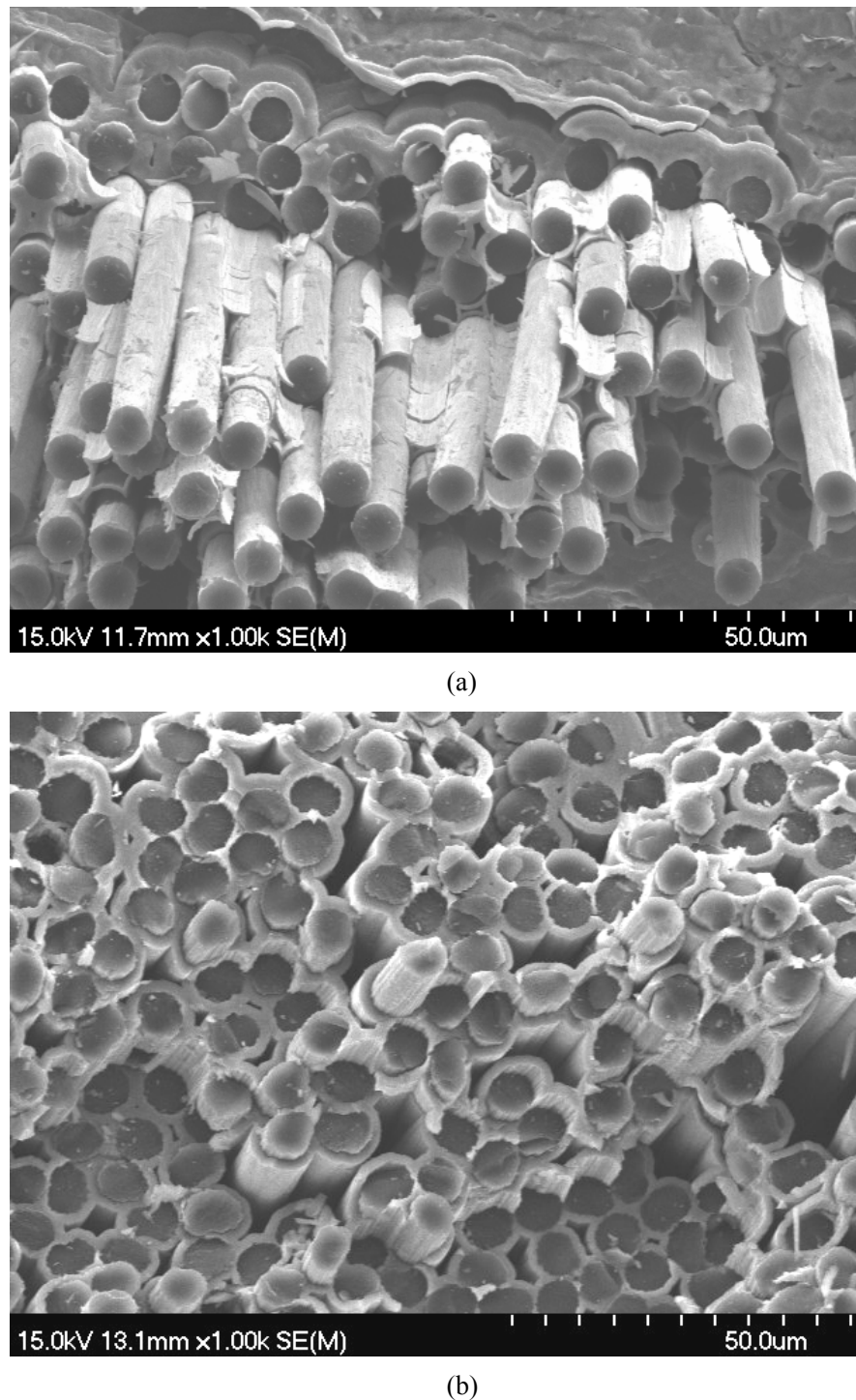


Fig.4: SEM fractographs of tensile specimens show that 1k C/SiC exhibits extensive individual filaments pull-out compared to 3k C/SiC. (a) 1k C/SiC (b)3k C/SiC.

3.2.2 In-plane shear properties

The typical shear stress-strain curves of C/SiC composites with two different sizes of tows are shown in Fig.5. The stress is linear proportional to strain at low stress. In contrast to the tensile behaviour, the specimens continue to deform without rupture immediately when loaded to maximum, exhibiting

apparent yielding behaviour, which is reflected by a platform in the end of shear stress-strain curve.

The non-linearity in shear curves may results from two aspects. On one hand, the pore tips in the material enable a large stress concentration when the material is loaded, which cause propagation of the

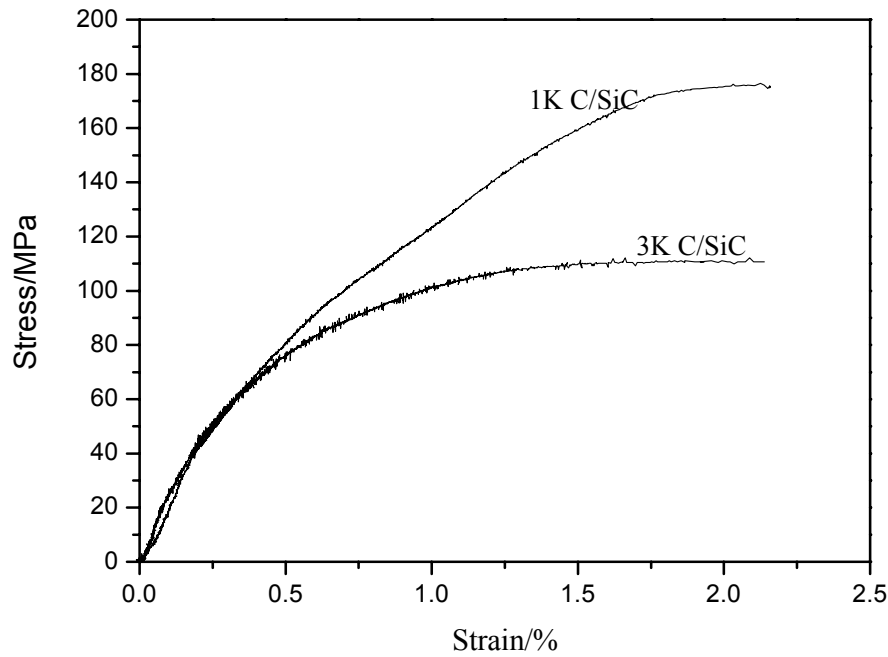


Fig.5: Typical shear stress-strain curves of plain woven C/SiC composites

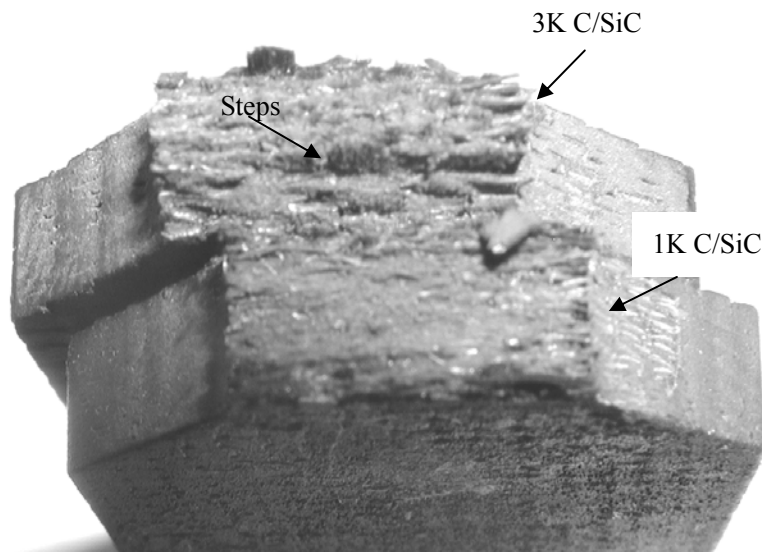


Fig.6: Macrofractographs of shear specimens show that the fracture surface of 3k C/SiC composite is far from planar than that of 1k C/SiC composite.

pre-existent microcracks, leading to a non-linearity in macro view. On the other hand, the fibre tows present shear deformation when specimen is shear loaded, and the interface between the intra-tow fibres and matrix is subjected to failure due to weak bonding. The interface debonding accumulates and develops to a large extent when loaded to maximum gradually, resulting in macro non-linearity.

The ultimate shear strength of 1k C/SiC composites is 28% higher than that of 3k C/SiC composites, although both the shear moduli of two composites are similar. Shear fracture occurs at the narrowest section of specimen, presenting a near-plane fracture surface. The macrofractographs of shear specimens are shown in Fig.6. It is observed that the fracture surface of 1k C/SiC specimen is flatter, while there

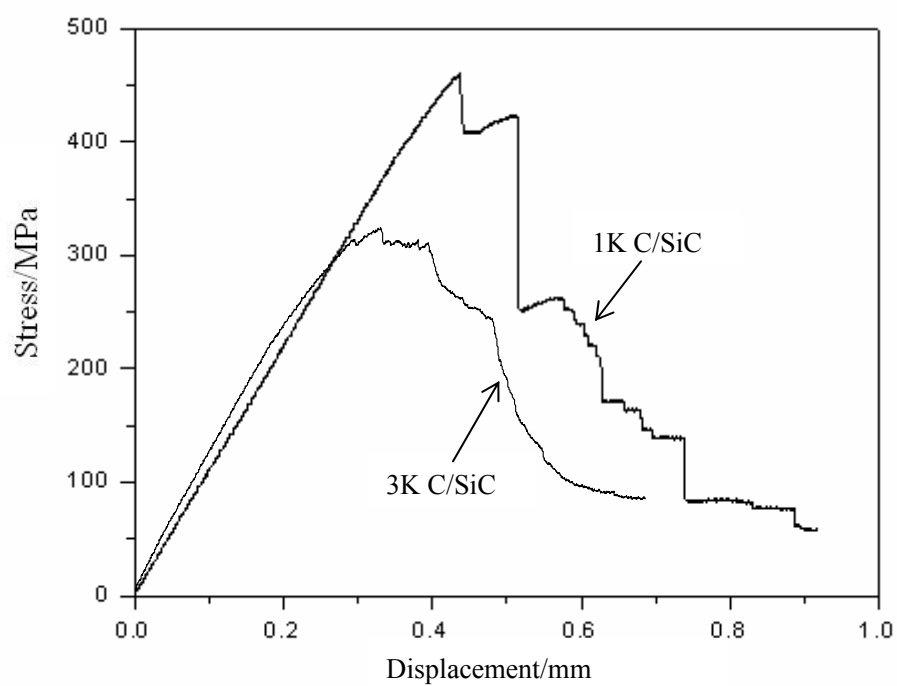


Fig.7: Typical flexural stress-displacement curves of plain woven C/SiC composites.

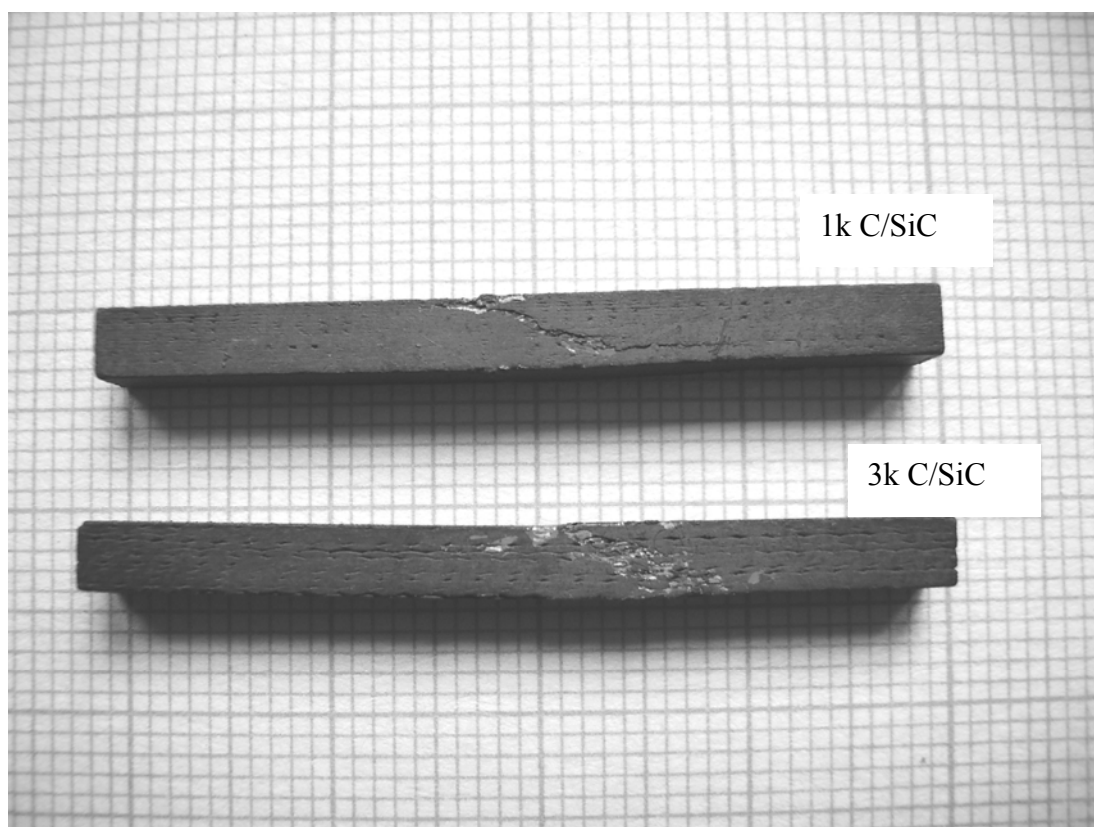


Fig.8: Macrographs of C/SiC composites after bending tests show that both composites exhibit delamination.

are many bulges or steps in fracture surface of 3k C/SiC specimen. Detailed microstructure observation revealed that 1k C/SiC specimen exhibits many individual filaments pull-out, while largely fibre bundles or clusters pull-out are the manifest characteristics of 3k C/SiC specimen, which is similar to the previous tensile specimens.

3.2.3 flexural properties

The typical flexural stress-displacement curves of C/SiC composites with two different sizes of tows are shown in Fig.7. It is observed that the flexural strength of 1k C/SiC composite is 11% higher than that of 3k C/SiC composite, but both the flexural modulus is close. The non-linearity feature of 3k C/SiC composite is evident compared with that of 1k C/SiC composite during the increase of load, and the stress decrease gradually after the maximum load. Therefore, both the composites exhibit a characteristic of non-catastrophic failure, or pseudo-plastic behaviour due to the matrix crack multiplication, which is most favourable to structure safety.

The macroscopic flexural fracture profile of C/SiC composites is shown in Fig.8. It is obvious that both the composites show delamination due to poor interlaminar shear strength, which is similar to the composites in tensile condition. The observation of fracture surface of the two composites reveals that 1k C/SiC composite illustrates a little fibre cluster pull-out and many filaments pull-out, while 3k C/SiC composite more clusters or bundles pull-out, indicating the undesirable bundle/matrix or bundle/bundle interfacial bonding, which is also similar to the previous tensile fracture surfaces.

4. CONCLUSIONS

Mechanical properties of two different 2D C/SiC composites subjected to tension, shear and bending have been described, suggesting that the comprehensive mechanical property of 1k C/SiC composite is superior to that of 3k C/SiC composite. The tensile strength, shear strength and bending strength of the 1 k C/SiC composites are 312 MPa, 163 MPa, 497 MPa, and are 58%, 28%, and 11% higher than that of the 3 k C/SiC composites, respectively. Both the

composites exhibit the mostly non-linear behaviour in the mechanical tests. However, SEM analysis of fracture surface reveal that the 1k C/SiC composite shows extensive filaments pull-out in contrast to the bundle or clusters pull-out in 3k C/SiC composite. The higher mechanical properties were attributed to the better degree of infiltration inside the tows and between plies, the smaller undulation angle, and desirable fibre/matrix or bundles/matrix interfacial bonding in 1k C/SiC composite.

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